

Devansh Parapalli

+91 8856962057 | Hyderabad, TG, India | hey@parapalli.dev | github.com/DevParapalli | linkedin.com/in/devparapalli | parapalli.dev

PROFESSIONAL SUMMARY

Solo engineer on the first production agent system at Cisco, now a technical lead of a team of 8 across support, finance, audit and automation projects. Owns AI systems end-to-end – requirements, architecture, data pipelines, production operations. 95% autonomous closure rate on support cases, 1,000+ person-hours saved per month. Published in IJTE (2025). GCP Professional Cloud Architect.

WORK EXPERIENCE

- Software Development Engineer - Technical Lead, AI Systems**

Jul 2025 — Present

Tata Consultancy Services Limited

Hyderabad, India

 - Client: Cisco** – First engineer on the production agent platform for Finance; grew to a team of 8 under my technical lead within 6 months. Owned requirements, architecture, and cross-functional alignment across support, FinOps, and audit teams over multiple concurrent production engagements.
 - Support automation:** Cut support triage from a full business day to under 30 seconds for multiple finance tracks within Cisco using multi-intent detection, tool-calling, MCP Servers, Kafka event streaming and auto-scaling runners. 95% of cases close autonomously, 4% complete AI-side pending human verification against external systems, 1% escalate unhandled; 100+ cases per track every month.
 - Finance automation:** Saved 1,000+ person-hours a month across payroll review and FinOps reporting by replacing manual work with AI-enabled pipelines over Snowflake, Oracle, MongoDB, plus a self-built DAG framework for extensible quarterly executive fund-flow reporting on highly restricted data; one quarter and one year-end closed with zero rework required.
 - Data reconciliation platform:** Designed and built a reconciliation engine that turns plain-English steps into concrete processing and matching operations, handling 100+ workflows at 1M+ rows per side with type-aware normalization, eliminating false-positives.
 - Audit validation:** Built an independent audit for a production document processing pipeline, validating its output through a deliberately different methodology and flagging where the two disagree; surfaces a real error in roughly 1 of every 30 documents the primary pipeline passed, at a false-flag rate of 1 in 1,000. Calibrated with eval sets and human review, sampling 30% of global invoices daily. Replaced an LLM-as-a-Judge prototype for speed and cost reasons.
 - Internal: GenAI Center of Excellence** – Led technical architecture and client demos for 10 RFPs spanning GenAI enablement, cloud migration, and legacy modernization.
 - Architected an on-premises vLLM + LiteLLM platform with gateway routing, PII detection, RBAC, and audit logging for internal model serving with no external API dependency; handed to a platform team for production.
 - Authored a GenAI adoption plan for a European semiconductor manufacturer covering 21 different use cases on Vertex AI and Gemini Enterprise, including on-premises infrastructure so confidential engineering data never leaves their data center.
 - Architected a replacement for a global telecom equipment maker’s manufacturing software to GKE, covering all production sites.

GenAI & Backend Development Intern

Jun 2024 — Aug 2024

PolymathAI

Nagpur, India

- Cut per-video multi-modal AI processing time from 3 minutes to 15 seconds by designing a parallelized inference pipeline across horizontally-scaled containers, processing 14,600+ videos in production.

PROJECTS

- AIKO: AI-powered Knowledge Organizer** (aiko.parapalli.dev)

Jun 2024 — Present
- Lifted Mean Reciprocal Rank from 0.25 to 0.75 and nDCG@5 from 0.5 to 0.7 over a 100K-chunk technical knowledge corpus by implementing hybrid retrieval (semantic search + keyword fallback) with tuned chunking and embedding selection across multi-modal content.
 - Sustained p95 ingest latency of 3s across a multi-modal document pipeline (PDFs, web content, multimedia) via a custom serialization layer with cloud-native and offline-capable deployment targets.
 - Built a custom orchestration framework for full control over prompt construction, retrieval, and tool-calling; replacing third-party abstractions with task-specific logic.
 - Reduced LLM response latency by 25% by replacing SSE with custom WebSocket duplex streaming for real-time token delivery.

EDUCATION

- Government College of Engineering, Nagpur**

Nagpur, India

B.Tech, Computer Science and Engineering

Dec 2021 — Jun 2025

CGPA: 8.28/10.0

TECHNICAL SKILLS

- AI & Agents:** LLM Serving (vLLM, LiteLLM), Multimodal Pipelines, Agentic Systems, Multi-Agent (A2A), MCP, LLM Evaluation, MLOps
- Retrieval:** RAG, Hybrid Retrieval, Embeddings, Vector DBs (pgvector, FAISS, Pinecone)
- Backend & Systems:** System Design, Distributed Systems, Event-Driven Architecture (Kafka), Async Concurrency, FastAPI
- Data:** Data Pipelines, Snowflake, PostgreSQL, Oracle, MongoDB, SQL
- Cloud & Security:** GCP (Certified Architect), AWS, Docker, Kubernetes, Terraform, OpenShift
- Languages:** Python, TypeScript, C/C++, SQL, PL/SQL

CERTIFICATIONS & ACHIEVEMENTS

- GCP Certifications:** Professional Cloud Architect (Dec 2025) | Cloud Digital Leader (Sep 2025) | Generative AI Leader (Sep 2025)
- Publication:** “AI-based Knowledge Organizer for diverse data formats” - IJTE, March 2025